Automatic Vehicle Number Plate Recognition Using Computer Vision Techniques

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology
in
Electronics and Communication Engineering

by

Tanmoy Mukherjee
Reg. No: 221300110696 of 2022-23 **Roll No:** 13000322088

Rupam Dutta
Reg. No: 221300111086 of 2022-23 **Roll No:** 13000322143

Piyas Mondal
Reg. No: 221300110632 of 2022-23 **Roll No:** 13000322123

Supriyo De
Reg. No: 221300110692 of 2022-23 **Roll No:** 13000322084

Under the Guidance of

Prof. Moumita Sarkar Kar

Department of Electronics and Communication Engineering

TECHNO MAIN SALT LAKE
EM 4/1 Salt Lake City, Sector-v, Kolkata-700091

2026

DECLARATION

We declare that the project report titled **"Automatic Vehicle Number Plate Recognition Using Computer Vision Techniques"** submitted in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in **Electronics and Communication Engineering** is a record of work carried out by us under the guidance of **Prof. Moumita Sarkar Kar** and has not formed the basis for the award of any other degree, in this or any other institution or university**.**

Date:

……………………………..
Tanmoy Mukherjee
Roll No - 13000322088

……………………………..
Supriyo De
Roll No - 13000322084

……………………………..
Rupam Dutta
Roll No – 13000322143

……………………………..
Piyas Mondal
Roll No - 13000322123

CERTIFICATE

This is to certify that Tanmoy Mukherjee, Rupam Dutta, Piyas Mondal & Supriyo De have carried out their final year project work entitled **"Automatic Vehicle Number Plate Recognition Using Computer Vision Techniques"** as a part of the curriculum for the B.Tech degree in Electronics & Communication Engineering (ECE) under Maulana Abul Kalam Azad University of Technology for the academic session 2025-2026.

This project report is approved by the undersigned only for the purpose for which it is submitted. The candidate is entirely responsible for the statements, opinions and conclusions contained herein.

Prof. Moumita Sarkar Kar
Assistant Professor
Dept. of ECE, TMSL

Dr. Arpita Adhikari
Assistant Professor & Head
Dept. of ECE, TMSL

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Date:

……………………………..
Tanmoy Mukherjee
Roll No - 13000322088

……………………………..
Supriyo De
Roll No - 13000322084

……………………………..
Rupam Dutta
Roll No – 13000322143

……………………………..
Piyas Mondal
Roll No - 13000322123

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ABSTRACT

The proposed project, Solar-Powered Automatic Number Plate Recognition (ANPR) System, aims to develop an intelligent, energy-efficient, and autonomous vehicle identification system using computer vision techniques. The system is designed to automatically detect, extract, and recognize vehicle registration numbers from images captured by an infrared (IR) camera during both daytime and nighttime conditions. A Raspberry Pi 4 serves as the central processing unit, executing image processing and Optical Character Recognition (OCR) algorithms using Python, OpenCV, and Tesseract OCR.

The methodology involves capturing vehicle images, preprocessing them through grayscale conversion, noise reduction, and edge detection, followed by number plate localization using contour analysis. The detected plate region is then segmented and passed to the OCR engine to extract alphanumeric characters. The recognized vehicle number, along with the date and time, is stored locally for future analysis.

To ensure operation in remote and power-deficient locations, the entire system is powered through a solar photovoltaic panel, charge controller, battery storage unit, and DC-DC converter, making it independent of grid electricity. The project provides a low-cost, offline, and sustainable solution for traffic monitoring, security surveillance, parking management, and access control applications while demonstrating the practical integration of embedded systems, renewable energy, and computer vision technologies.

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CHAPTER 1: INTRODUCTION

1.1 General Overview

The transportation infrastructure of modern urban and semi-urban environments has undergone unprecedented expansion over the past two decades driven by rapid economic growth, increased purchasing power, and the widespread availability of affordable vehicles. In India alone, the total number of registered motor vehicles has surpassed 300 million, with metropolitan cities witnessing exponential growth in vehicular density. This surge in vehicular population has placed enormous strain on existing traffic management systems, law enforcement agencies, and surveillance infrastructure. Traditional methodologies for vehicle identification, which historically relied upon manual inspection and human vigilance at checkpoints, toll plazas, and restricted zones, have become increasingly impractical and inefficient in the face of such overwhelming volume.

The limitations of manual monitoring are manifold and well-documented. Human operators are inherently susceptible to fatigue, distraction, and cognitive overload, particularly during extended duty hours or in high-traffic scenarios. The identification of vehicles based on visual inspection of number plates is a time-consuming process that creates bottlenecks at critical junctions, leading to traffic congestion and operational delays. Furthermore, the enforcement of traffic regulations, detection of stolen vehicles, and surveillance of restricted areas demand a level of accuracy and speed that manual systems simply cannot provide. The dependence on human judgment introduces the risk of oversight, especially in challenging environmental conditions such as nighttime operations, heavy rainfall, or dense fog, where visibility is severely compromised [1].

In response to these challenges, automated vehicle identification systems have emerged as a critical technological intervention. Among these, automatic number plate recognition (ANPR) systems represent a sophisticated fusion of computer vision, image processing, and pattern recognition technologies. These systems are capable of capturing real-time images of moving vehicles, extracting the alphanumeric characters inscribed on registration plates, and converting this visual information into machine-readable digital data. The advantages of such automation are significant: ANPR systems operate continuously without fatigue, process information with high speed and accuracy, and generate structured databases that facilitate advanced analytics and investigative operations.

However, existing deployments face a persistent infrastructural constraint in the form of continuous electrical power supply. In regions characterized by intermittent power availability, rural highways, or remote checkpoints, conventional ANPR systems dependent on grid electricity experience frequent operational disruptions. This limitation undermines the reliability and utility of the technology, particularly in contexts where uninterrupted surveillance is paramount. The integration of renewable energy sources, specifically solar photovoltaic systems, offers a viable solution to this challenge. Solar-powered ANPR systems can operate autonomously, independent of grid infrastructure, ensuring continuous functionality even in off-grid or power-scarce environments.

This project, titled "automatic vehicle number plate recognition using computer vision technique with solar power," proposes the design, development, and validation of a fully functional prototype that integrates state-of-the-art computer vision algorithms with sustainable energy infrastructure. The system leverages the computational capabilities of a raspberry pi 4 single-board computer, the imaging precision of a dedicated camera module, and the energy independence afforded by a solar panel with battery backup. By combining these elements, the proposed system aims to deliver a robust, cost-effective, and environmentally sustainable solution for automated vehicle identification, suitable for deployment across diverse geographical and infrastructural contexts.

1.2 Evolution of Number Plate Recognition Technology

The evolution of vehicle identification technology is a narrative that mirrors the broader trajectory of automation and digitization in public infrastructure. In the earliest phases of motorized transportation, vehicle identification was an entirely manual endeavour. Traffic police and checkpoint personnel relied upon direct visual inspection of registration plates, cross-referencing observed numbers against handwritten or printed registers. This method, while functional in an era of limited vehicular density, was inherently slow, error-prone, and incapable of scaling to accommodate growing traffic volumes.

The advent of closed-circuit television (CCTV) systems in the late 20th century marked the first significant technological advancement in vehicle surveillance. CCTV installations at strategic locations enabled the recording of vehicle movements for subsequent analysis. However, these systems were fundamentally passive; they generated vast quantities of video footage that required human operators to review manually. The identification of specific vehicles within hours of recorded content remained a labour-intensive task, often undertaken only in response to specific incidents or investigative requirements. Real-time identification was still unattainable, and the utility of CCTV for proactive traffic management or law enforcement was limited [2].

The introduction of digital image processing techniques in the early 2000s represented a paradigm shift. Researchers and engineers began exploring the possibility of automating the extraction of textual information from video frames. Early ANPR prototypes employed rudimentary edge detection algorithms and template matching methods to isolate rectangular regions likely to contain number plates. Optical character recognition (OCR) engines, originally developed for document digitization, were adapted to interpret the alphanumeric characters present in these extracted regions. These pioneering systems demonstrated proof-of-concept but suffered from poor accuracy, particularly under variable lighting conditions, oblique viewing angles, or when plates were obscured by dirt and debris.

The maturation of computer vision as a discipline, coupled with exponential increases in computational power and the proliferation of high-resolution digital cameras, has enabled the development of modern ANPR systems characterized by remarkable sophistication. Contemporary implementations utilize machine learning algorithms, including convolutional neural networks (CNNS), to achieve highly accurate localization of number plates even in complex visual scenes. Advanced preprocessing techniques compensate for variations in illumination, perspective distortion, and motion blur. Real-time processing capabilities allow these systems to analyse video streams at frame rates exceeding thirty frames per second, enabling instantaneous identification of vehicles in motion.

Despite these advancements, the deployment of ANPR technology has been predominantly confined to environments with reliable electrical infrastructure. Urban toll plazas, airport parking facilities, and metropolitan traffic intersections typically possess the necessary power supply and network connectivity to support conventional ANPR installations. However, the extension of these capabilities to rural highways, border checkpoints, or temporary surveillance locations has been hindered by the unavailability of consistent grid power. This infrastructural gap underscores the necessity for energy-autonomous ANPR systems, a requirement that contemporary research is only beginning to address systematically.

1.3 Problem Identification and Analysis

A comprehensive analysis of existing vehicle identification methodologies, informed by both academic literature and ground-level operational observations, has revealed several critical deficiencies that serve as the impetus for the proposed system. These problems are not merely theoretical concerns but represent tangible barriers to the effective deployment and utilization of automated vehicle surveillance technologies, particularly in the diverse and challenging contexts characteristic of indian infrastructure.

The first and most fundamental problem is the inherent fallibility and inefficiency of manual vehicle monitoring. Human operators tasked with recording vehicle registration numbers at checkpoints or toll booths are subject to a multitude of cognitive and physiological limitations. Sustained attention over extended periods leads to mental fatigue, resulting in decreased vigilance and increased error rates. In high-traffic scenarios, where vehicles pass in rapid succession, the task of accurately transcribing alphanumeric sequences becomes cognitively demanding and prone to mistakes. Misidentification of characters due to similar visual appearance, such as confusing the letter 'o' with the digit '0' or the letter 'b' with the digit '8', is a recurrent issue. Furthermore, deliberate evasion tactics, such as obscured plates or high-speed passage, can easily circumvent manual surveillance. The operational cost of maintaining round-the-clock human personnel at multiple locations is also economically prohibitive, particularly for resource-constrained agencies [3].

The second significant challenge pertains to operational reliability under adverse environmental conditions. Nighttime operations present a particularly acute problem. The reflective properties of standard vehicle number plates are optimized for daylight visibility, but under low ambient illumination, these plates become difficult to discern with the naked eye or with standard visible-spectrum cameras. Traditional CCTV systems, which rely on ambient or artificial white light, produce images of insufficient quality during nighttime, rendering manual identification nearly impossible. Similarly, adverse weather conditions such as heavy rainfall, fog, or dust storms degrade visibility and image quality, further compromising the effectiveness of both manual and rudimentary automated systems. The absence of specialized illumination and imaging solutions tailored for these conditions represents a significant gap in existing surveillance infrastructure.

The third critical problem is the dependency of conventional ANPR systems on uninterrupted electrical power supply from the grid. Many rural highways, remote border crossings, and temporary checkpoints lack reliable access to consistent electricity. Even in areas nominally connected to the grid, frequent power outages and voltage fluctuations are common. Conventional ANPR installations, which typically comprise power-hungry components such as desktop computers, external lighting arrays, and network routers, cannot function during power failures. The use of diesel generators as backup power sources is both economically and environmentally unsustainable, entailing high fuel costs, maintenance overhead, and carbon emissions. This power dependency fundamentally limits the geographic and operational scope of ANPR technology, confining ITS deployment to well-serviced urban locations and excluding vast swathes of territory where surveillance is equally necessary.

The fourth problem concerns the difficulty of achieving real-time identification and response. In time-critical scenarios, such as the pursuit of vehicles involved in criminal activities or the enforcement of access control at high-security facilities, delays in vehicle identification can have serious consequences. Manual systems inherently operate with significant latency; the time required for an operator to visually observe a plate, transcribe the number, and cross-reference it against a database can extend to several minutes. Even first-generation automated systems, burdened by slow processing algorithms and inadequate computational hardware, often fail to deliver results with the immediacy required for actionable intervention.

These problems collectively underscore the necessity for a technological solution that is accurate, autonomous, environmentally resilient, energy-independent, and capable of real-time operation. The proposed system directly addresses each of these identified deficiencies through the integration of advanced computer vision techniques, specialized imaging hardware, and renewable solar energy infrastructure.

1.4 Motivation

The motivation for undertaking this project stems from a convergence of technological opportunity, societal need, and environmental responsibility. The contemporary landscape of traffic management and law enforcement is characterized by an urgent requirement for intelligent automation systems capable of augmenting human capabilities and compensating for infrastructural limitations. The specific motivation for developing a solar-powered ANPR system is rooted in several compelling factors that extend beyond mere academic inquiry to encompass practical, operational, and ethical dimensions.

Foremost among these motivations is the imperative to automate vehicle identification with a level of reliability and consistency unattainable through manual means. Automated systems eliminate the variability introduced by human factors, ensuring that every vehicle passing within the system's field of view is captured, analysed, and recorded with uniform diligence. This consistency is crucial for applications such as toll collection, parking management, traffic flow analysis, and security surveillance.

A second significant motivation is the enhancement of traffic law enforcement capabilities. Violations such as speeding, unauthorized entry into restricted zones, and the use of unregistered or stolen vehicles pose ongoing challenges to law enforcement agencies. An automated ANPR system, capable of continuous operation and equipped with database integration, can instantaneously flag vehicles of interest, enabling rapid response and intervention. The deterrent effect of pervasive automated surveillance is well-documented; the knowledge that violations will be detected and recorded discourages non-compliance and contributes to overall road safety.

The third motivation concerns operational resilience under diverse environmental conditions. The ability to function effectively during nighttime hours and adverse weather is not a luxury but a necessity for comprehensive surveillance. By incorporating Infrared imaging technology and specialized illumination, the proposed system ensures that vehicle identification capability is maintained regardless of ambient lighting conditions [4].

The fourth and perhaps most forward-looking motivation is the integration of renewable solar energy to achieve energy independence. The environmental and economic sustainability of technological infrastructure is an increasingly critical consideration in the context of global climate change and resource constraints. Solar photovoltaic technology has matured to the point where small-scale, off-grid power systems are both technically viable and economically feasible. By harnessing solar energy, the proposed system eliminates recurring electricity costs, reduces carbon footprint, and enables deployment in locations where grid connectivity is absent or unreliable.

1.5 Objectives of The Project

The primary objective of this project is to design, develop, and validate a fully functional prototype of an automatic number plate recognition system powered by solar energy. The specific technical and operational objectives are as follows:

- To achieve automatic detection and localization of vehicle number plates from real-time video streams captured by a dedicated camera module, employing computer vision algorithms capable of isolating the rectangular region corresponding to the registration plate even in visually complex scenes.
- To implement accurate character recognition using optical character recognition techniques, enabling the extraction and interpretation of alphanumeric sequences from the detected number plate region, with the capability to handle variations in font, spacing, and minor occlusions.

- To ensure reliable operation during both day and night conditions through the integration of Infrared illumination and appropriate image preprocessing techniques, thereby maintaining consistent identification accuracy irrespective of ambient lighting levels.
- To design and implement a solar-powered energy system comprising photovoltaic panels and rechargeable battery storage, ensuring autonomous operation of the entire ANPR hardware without dependence on grid electricity.
- To develop a data storage and retrieval mechanism for recording identified number plates along with timestamps and captured images, facilitating the creation of a structured database for subsequent analysis or integration with law enforcement databases.
- To validate the system's performance through comprehensive testing under varying environmental conditions, including different times of day, weather scenarios, and vehicle speeds, quantifying accuracy metrics such as detection rate, recognition accuracy, and processing latency.
- To ensure that the system is cost-effective and suitable for replication and deployment in real-world scenarios, utilizing widely available hardware components and open-source software frameworks.

1.6 Scope and Constraints

The scope of this project encompasses the design, construction, and testing of a prototype-level automatic number plate recognition system integrated with a solar power supply. The system is engineered to operate as a standalone, offline module capable of capturing vehicle images, detecting and extracting number plates, performing character recognition, and storing the resultant data locally on the processing unit.

The system operates entirely offline, meaning that it does not require internet connectivity or access to external cloud-based services during normal operation. All image processing, character recognition, and data storage functions are executed locally on the raspberry pi 4 single-board computer. The captured data, including recognized number plate strings and associated images, is stored on the microsd card within the raspberry pi and can be periodically retrieved for analysis or archival purposes.

However, the project operates under certain constraints that must be acknowledged. Firstly, the system is designed to recognize number plates conforming to the standard format prescribed by indian vehicle registration authorities. Secondly, the accuracy of the system is contingent upon certain environmental and operational assumptions; the camera must be positioned at an appropriate distance and angle relative to the vehicle to capture a clear image of the number plate.

Thirdly, the solar power system is dimensioned to support continuous operation under typical solar irradiance conditions. However, extended periods of overcast weather or inadequate battery capacity could potentially lead to power depletion. The current prototype uses a battery with capacity sufficient for approximately twenty-four to forty-eight hours of autonomous operation without solar charging.

Fourthly, the computational performance of the raspberry pi 4, while adequate for real-time processing of single-lane traffic at moderate speeds, may encounter limitations in scenarios involving multiple simultaneous vehicles or very high frame rates. The processing latency is approximately one to two seconds per vehicle, which is acceptable for checkpoint applications.

Finally, the project assumes that the enrolment of vehicle number plates into a reference database is conducted as a pre-deployment activity. The system does not currently implement live synchronization with central government databases such as vahan, though such integration is technically feasible and represents a logical avenue for future enhancement.

1.7 Organization of The Report

The subsequent chapters of this report are structured to provide a comprehensive and systematic exposition of the project's development, adhering to standard practices for engineering documentation and academic reporting.

Chapter 2, titled literature review, presents a critical survey of contemporary research in the domain of automatic number plate recognition and related computer vision applications. The literature review identifies specific technological gaps, particularly the scarcity of low-cost, solar-powered ANPR systems optimized for offline operation, thereby establishing the novelty and relevance of the proposed project.

Chapter 3, titled methodology, elucidates the comprehensive design and implementation strategy adopted for the development of the proposed system. This chapter commences with a high-level overview of the system architecture, followed by detailed functional block diagrams, hardware component selection, software design, and a discussion of system integration challenges.

Chapter 4, titled resource utilization, documents the component-wise bill of materials and cost analysis for the proposed prototype system.

Chapter 5, titled measurements, results and discussion, documents the practical realization of the prototype and presents an evaluation of ITS performance. Comprehensive testing protocols are detailed, encompassing experiments conducted under varied lighting conditions, different vehicle speeds, and diverse environmental scenarios.

Chapter 6, titled conclusion and future scope, synthesizes the findings and achievements of the project, evaluating the extent to which the stated objectives have been realized. It also identifies residual limitations and proposes avenues for future enhancement.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rapid proliferation of vehicular traffic in urban and rural landscapes has catalysed the evolution of intelligent transportation systems (ITS) as a critical domain of research and development. Intelligent transportation systems encompass a wide spectrum of technologies designed to enhance the safety, efficiency, and sustainability of transportation networks through the integration of sensors, communication systems, data analytics, and automated decision-making mechanisms [5]. Among the various components of ITS, vehicle surveillance and automated identification systems occupy a position of paramount importance, serving as the foundational infrastructure for traffic monitoring, law enforcement, toll collection, parking management, and security applications .

Automatic number plate recognition (ANPR) systems represent a sophisticated application of computer vision and pattern recognition technologies, enabling the automated extraction and interpretation of alphanumeric characters inscribed on vehicle registration plates. The significance of ANPR technology has intensified over the past decade, driven by the exponential growth in vehicle population, the increasing complexity of urban traffic networks, and the limitations inherent in manual surveillance methodologies [6].

This chapter presents a comprehensive review of contemporary research in the domain of automatic number plate recognition, with particular emphasis on works published during the period spanning 2019 to 2024. The review is organized thematically, examining classical computer vision techniques, machine learning innovations, and embedded system implementations. Special attention is devoted to the exploration of energy-efficient and solar-powered surveillance systems, a domain of increasing relevance in the context of sustainable infrastructure development.

2.2 Computer Vision Based ANPR Systems (2019–2021)

The foundational methodologies employed in automatic number plate recognition systems during the period from 2019 to 2021 were predominantly rooted in classical computer vision techniques, which leveraged image processing algorithms to detect, localize, and recognize vehicle registration plates. These approaches, while computationally efficient and suitable for deployment on resource-constrained hardware, exhibited characteristic strengths and limitations that shaped subsequent research directions [6].

A significant body of research during this period focused on the application of edge detection algorithms as the primary mechanism for number plate localization. Edge detection techniques, including the sobel operator, canny edge detector, and prewitt filters, were employed to identify regions of high gradient intensity within captured images, corresponding to the boundaries of rectangular number plates [6]. Researchers demonstrated that by applying appropriate threshold values and combining horizontal and vertical edge maps, it was possible to isolate candidate regions likely to contain registration plates. Following edge detection, morphological operations such as dilation, erosion, opening, and closing were systematically applied to refine the detected regions [5].

Contour analysis emerged as a crucial subsequent step in the processing pipeline. After morphological refinement, connected component analysis was performed to identify distinct contours within the processed image. Geometric filters were applied to these contours, retaining only those regions that satisfied criteria consistent with the expected dimensions and aspect ratio of standard number plates [6].

Once candidate plate regions were successfully localized, optical character recognition (OCR) techniques were employed to interpret the alphanumeric content. Classical OCR approaches utilized template matching, wherein each segmented character was compared against a pre-stored library of character templates [5]. Several research implementations reported recognition accuracy rates exceeding ninety percent under controlled conditions, validating the feasibility of classical computer vision approaches for ANPR applications [6].

However, critical analysis of these classical methodologies revealed several significant limitations that constrained their practical utility in real-world deployments. The most pronounced deficiency was sensitivity to variations in illumination [7]. Edge detection algorithms exhibited degraded performance under conditions of extreme brightness, deep shadows, or non-uniform lighting. Nighttime operation posed substantial challenges, as the absence of adequate ambient illumination resulted in low-contrast images from which edges could not be reliably extracted [6].

Variations in viewing angle constituted another major source of error. Number plates captured at oblique angles exhibited perspective distortion, altering the apparent aspect ratio and introducing non-uniform character spacing [4][5]. Environmental factors such as weather conditions further exacerbated the limitations of classical approaches. Heavy rainfall, fog, and dust obscured visibility, attenuating image contrast and introducing noise that interfered with edge detection [6]. Collectively, these limitations underscored the vulnerability of classical computer vision techniques to uncontrolled environmental variability [3][7].

2.3 Machine Learning and Advanced ANPR Approaches (2021–2023)

The period from 2021 to 2023 witnessed a pronounced shift in ANPR research toward the adoption of machine learning techniques, particularly deep learning architectures employing convolutional neural networks (CNNs). This transition was driven by the recognition that handcrafted feature extraction methods, inherent to classical computer vision, were insufficiently robust to accommodate the diversity and complexity of real-world imaging conditions [8].

Convolutional neural networks emerged as the dominant paradigm for number plate detection and localization. Researchers employed architectures inspired by object detection frameworks, wherein the network was trained to identify the spatial coordinates of bounding boxes enclosing number plates within input images [6]. These models processed images through successive convolutional and pooling layers, extracting hierarchical feature maps that captured increasingly abstract representations of visual patterns [7]. Studies reported substantial improvements in detection accuracy compared to classical methods, with trained networks demonstrating resilience to challenging conditions such as partial occlusions, varying fonts, and non-standard plate designs.

Character recognition also benefited significantly from the application of deep learning. Rather than relying upon template matching or hand-engineered features, researchers trained

CNNs to directly classify segmented character images into corresponding alphanumeric categories [8]. Some research implementations reported character-level accuracies exceeding ninety-five percent, even in the presence of moderate noise or distortion [6][7].

Advanced research during this period also explored end-to-end trainable architectures that integrated detection and recognition into a unified framework. These systems eliminated the need for explicit character segmentation, instead directly predicting character sequences from detected plate regions using recurrent neural network components or attention mechanisms [8]. The end-to-end approach simplified the processing pipeline and reduced cumulative error propagation.

Despite these impressive advancements, the adoption of machine learning-based ANPR systems introduced a new set of challenges. The most critical limitation pertained to computational cost [9]. Deep learning models demanded substantial computational resources for both training and inference. The execution of real-time inference on high-resolution video streams required processing hardware equipped with powerful multi-core processors or dedicated graphics processing units. Research works acknowledged that deployment on embedded platforms with limited computational capacity was impractical without significant architectural simplifications or model compression techniques [6][8].

The dependency on powerful hardware translated directly into elevated power consumption, a consideration of critical importance for autonomous or solar-powered installations [7]. Graphics processing units and high-performance processors consume tens to hundreds of watts of electrical power during sustained operation. In off-grid or remote locations where solar energy represented the sole available power source, the energy demands of machine learning-based systems posed significant challenges to continuous operation [9].

2.4 Embedded and Solar-Powered Surveillance Systems (2022–2024)

The most recent phase of research, spanning the years 2022 to 2024, has been characterized by a growing emphasis on embedded system implementations and the integration of renewable energy sources. This research direction reflects a pragmatic recognition of the infrastructural realities confronting the deployment of intelligent transportation systems in developing regions, rural environments, and temporary monitoring installations [10].

Embedded ANPR systems, built upon compact single-board computers and microcontrollers, have garnered considerable research attention as viable alternatives to conventional workstation-based architectures. Research implementations demonstrated that by employing optimized algorithms, efficient programming practices, and judicious selection of hardware peripherals, it was possible to construct functional ANPR prototypes capable of processing video streams at acceptable frame rates [9][10].

The integration of solar photovoltaic technology with surveillance systems emerged as a prominent research theme during this period. The motivation for solar integration was rooted in both environmental sustainability considerations and practical operational necessities [12]. Research works explored various configurations of solar panels, charge controllers, and battery storage systems, seeking to optimize the balance between energy generation, storage capacity, and system load [11].

Investigations into solar-powered surveillance systems revealed that careful energy management was essential to ensure uninterrupted operation. The intermittent nature of solar irradiance, subject to diurnal cycles and weather variability, necessitated the incorporation of energy storage mechanisms capable of sustaining system operation during periods of low or absent sunlight [10][12]. Researchers analysed battery technologies, comparing lead-acid, lithium-ion, and lithium-polymer chemistries with respect to energy density, cycle life, and temperature tolerance.

Environmental robustness emerged as a recurring challenge in the design of outdoor solar-powered systems. Exposure to temperature extremes, humidity, precipitation, and dust necessitated the implementation of protective enclosures and thermal management strategies [10]. Despite these innovations, several research gaps persisted. The majority of solar-powered surveillance systems documented in the literature were designed for relatively simple tasks such as motion detection or low-resolution image capture [10]. Comprehensive ANPR implementations that integrated high-resolution imaging, real-time processing, and extended autonomous operation remained scarce [11][12].

2.5 Synthesis and Research Gap Identification

A systematic synthesis of the reviewed literature reveals a fragmented landscape of ANPR research wherein individual studies have addressed specific dimensions of the problem, but few have achieved a holistic integration of these considerations [13]. Five principal research gaps are identified.

The first and most salient research gap pertains to the scarcity of low-cost, solar-powered ANPR systems that integrate all functional requirements into a unified, deployable solution. Existing high-accuracy ANPR systems typically assume the availability of grid power and high-performance computing hardware, rendering them unsuitable for off-grid deployment [14].

The second critical gap concerns the emphasis on cloud-based or network-dependent architectures in contemporary ANPR research. In rural and remote regions, where internet connectivity is intermittent or entirely absent, cloud-dependent systems are impractical. The development of fully offline, standalone ANPR systems capable of autonomous operation remains an under addressed challenge [14].

The third gap relates to the operational performance of ANPR systems in power-scarce and infrastructurally challenged environments. Relatively little attention has been devoted to the unique challenges posed by rural highways, forest checkpoints, or temporary installations in developing regions [13][15].

The fourth identified gap concerns the over-reliance on high-end hardware in recent machine learning-based approaches. Research exploring the optimization of machine learning models for embedded execution has made progress, but practical implementations demonstrating real-time ANPR performance on resource-constrained hardware powered entirely by solar energy remain rare [14].

The fifth gap pertains to nighttime operation and the integration of Infrared imaging technology. Research providing detailed design guidance, performance characterization, and validation of Infrared-based ANPR systems suitable for embedded, solar-powered deployment is notably absent [13][14].

2.6 Summary

This chapter has presented a comprehensive review of contemporary research in automatic number plate recognition, tracing the evolution of methodologies from classical computer vision approaches through machine learning innovations to embedded and solar-powered implementations. Classical computer vision techniques demonstrated computational efficiency but exhibited vulnerability to variations in illumination, perspective, and environmental conditions. Machine learning approaches delivered superior accuracy but imposed substantial computational and energy demands that rendered them impractical for embedded, solar-powered deployment [16][17].

The synthesis of reviewed works has identified five principal research gaps: the absence of integrated low-cost solar-powered ANPR systems; over-reliance on network connectivity and cloud processing; lack of validation in power-scarce rural environments; dependency on high-end computational hardware; and insufficient attention to nighttime operation using Infrared technology [16]. These gaps establish the rationale and motivation for the proposed project, which seeks to develop a holistic solution that addresses cost, energy independence, operational resilience, and deployment accessibility simultaneously [17].

CHAPTER 3: METHODOLOGY

3.1 Overview of the Proposed System

The proposed system is designed as an offline, standalone embedded platform for automatic vehicle number plate recognition using computer vision techniques powered by a dedicated solar energy subsystem. It integrates a processing unit, an imaging unit, an illumination module, and a solar power supply unit into a compact, field-deployable setup suitable for roadside or access-control installations. The core functionality of the system is to continuously monitor an approach lane, capture images of passing vehicles, detect and localize the number plate, and perform character recognition to extract the registration number for further processing or storage.

The system architecture follows a modular design approach, where each subsystem is responsible for a specific function such as image acquisition, preprocessing, plate detection, character segmentation, recognition, and data logging. The imaging unit consists of a camera module oriented towards the vehicle lane, assisted by Infrared illumination to support low-light and night-time operation without causing glare or driver distraction.

3.2 Functional Block Diagram Description

The functional block diagram of the proposed system provides a high-level view of the main modules and the flow of signals and data between them, as shown in fig. 3.1.

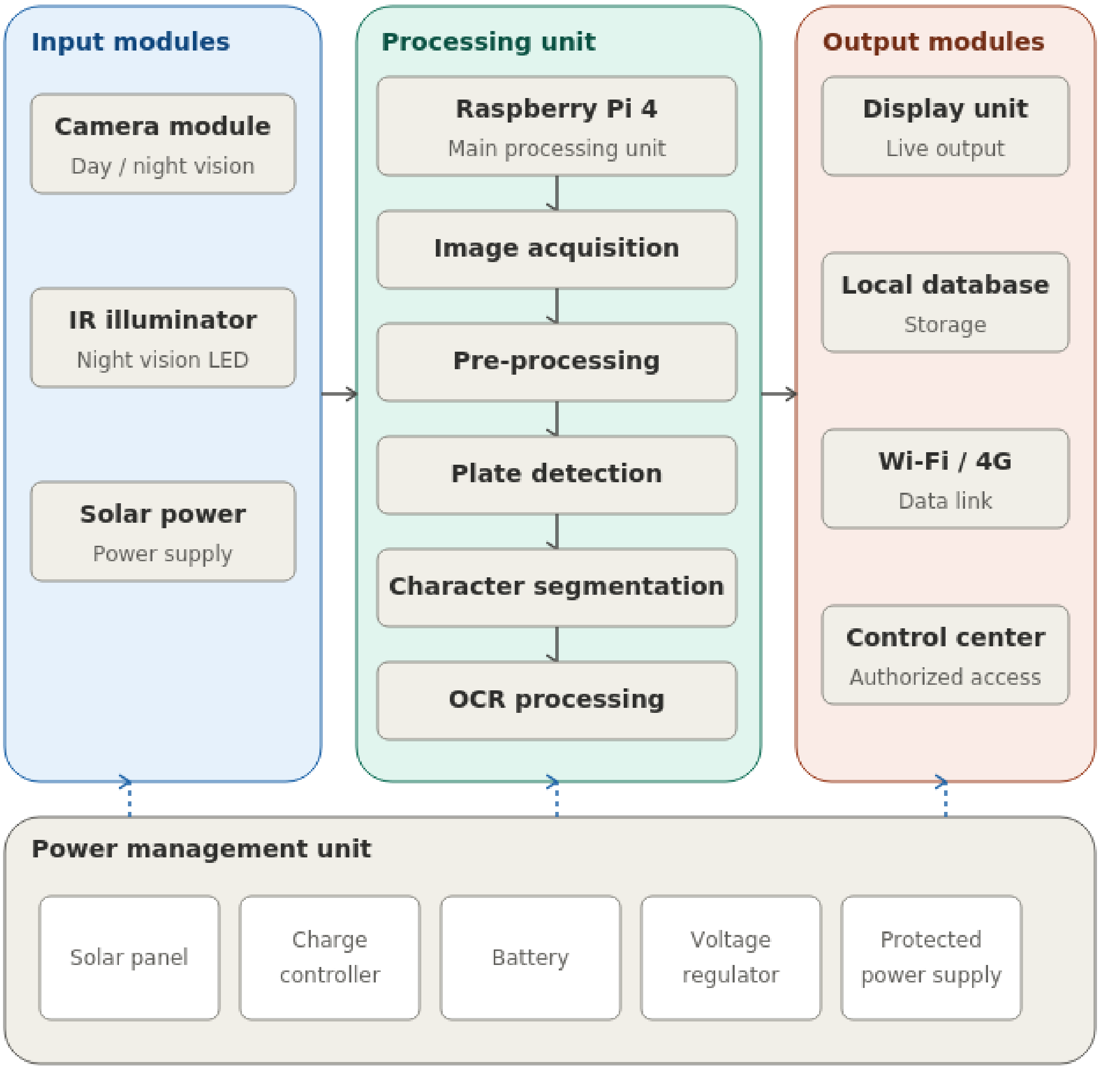

Fig. 3.1. Functional block diagram of the system

At the input side, the system comprises the camera module, Infrared (IR) illumination unit, and solar power supply unit. The camera module is responsible for capturing images or frames of the vehicles as they pass within the field of view. The Infrared (IR) illumination unit is synchronized with the camera to enhance visibility of the number plate in low-light conditions. The solar power supply unit converts incident solar energy into electrical power and stores it in a battery.

The central block is the processing unit, which acts as the computational core of the system. It receives image data from the camera module and executes the complete ANPR pipeline, which includes image acquisition, preprocessing, plate localization, character segmentation, and OCR-based recognition. The processing unit also manages system control tasks, such as scheduling capture events, monitoring power status, and controlling the Infrared (IR) illumination according to ambient light conditions.

At the output side, the recognized number plate string is stored in a local database or log file, along with optional metadata such as date, time, and capture status. A local display or interface may be used to show the recognized plate or system status for verification and debugging.

3.3 Hardware System Design and Component Selection

The processing unit is selected to provide sufficient computational resources for real-time or near real-time execution of the ANPR algorithm, while maintaining low power consumption suitable for solar-powered operation. It also features interfaces for the camera module, storage, and general-purpose input-output pins for controlling peripherals such as the Infrared (IR) illumination and power monitoring circuitry.

The camera module is mounted in a fixed orientation facing the vehicle lane, with an appropriate focal length and field of view to ensure that the number plate occupies a sufficient portion of the captured frame at the typical capture distance. The camera is enclosed in a protective housing to guard against dust, rain, and accidental impact. The Infrared (IR) illumination unit is placed near the camera, oriented in the same direction, so that the number plate area is uniformly illuminated during low ambient light conditions.

The solar power subsystem forms a critical part of the hardware architecture and is shown in detail in fig. 3.3. This subsystem consists of a solar panel, a charge controller, and a rechargeable battery. The solar panel converts solar energy into electrical power, which is regulated by the charge controller to safely charge the battery and protect it from overcharging and deep discharge.

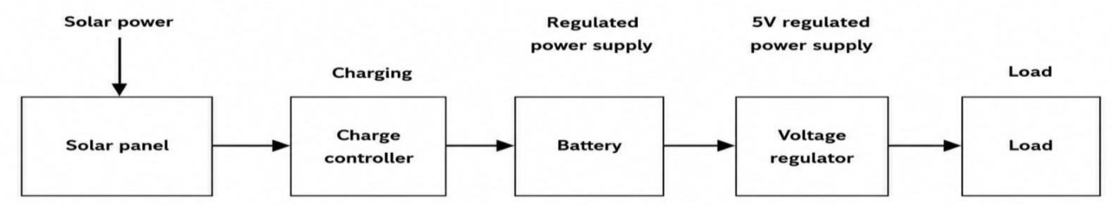

Fig. 3.2. Solar power supply unit

Fig. 3.4 shows the GPIO pin configuration of raspberry pi 4 used in the proposed system. GPIO pins are utilized for controlling Infrared (IR) illumination, monitoring power status signals, and interfacing with auxiliary peripherals.

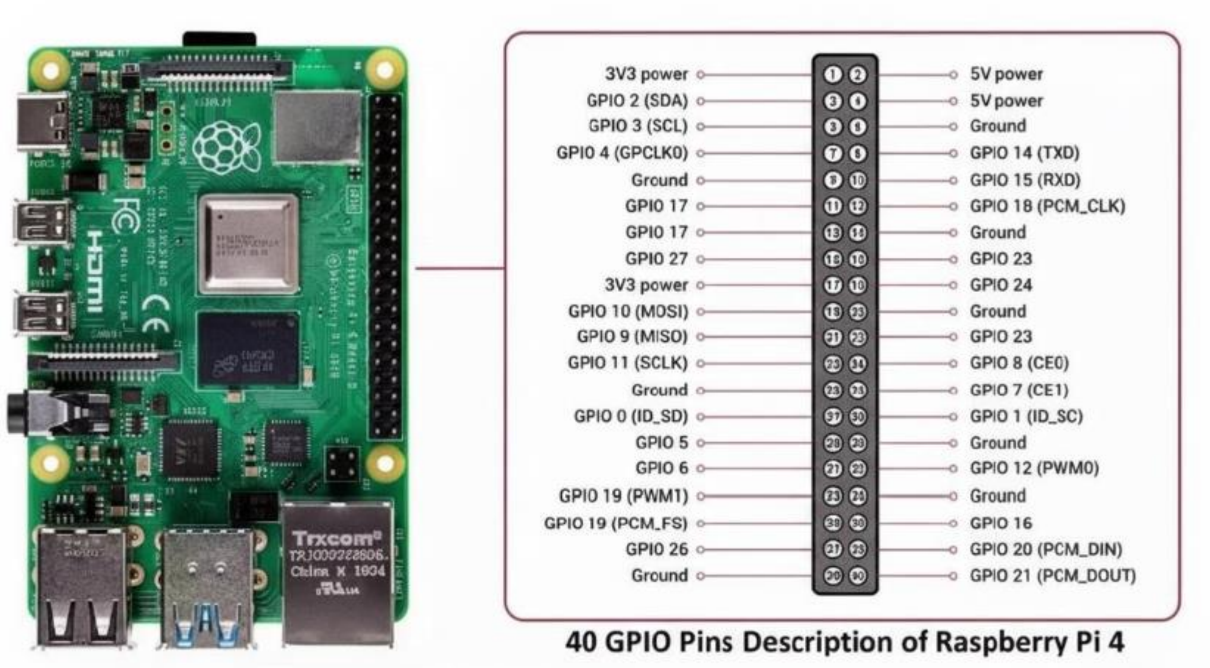

Fig. 3.3. Raspberry pi 4 pin diagram

3.4 Software Design and Algorithm Implementation

The software architecture is structured as a sequence of clearly defined stages that collectively implement the ANPR functionality on the embedded processing platform. The high-level flow of the algorithm is depicted in fig. 3.5.

The process begins with system initialization, where hardware interfaces such as the camera, storage and any required parameters for image processing are loaded. Once initialization is complete, the system enters a continuous monitoring loop, during which frames are periodically captured from the camera based on a predefined frame rate or event trigger.

The captured frames are first subjected to preprocessing, which typically includes resizing, conversion to grayscale, noise reduction, and contrast enhancement to improve the visibility of edges and characters on the number plate. Following preprocessing, the plate localization module attempts to detect the region of interest corresponding to the number plate. This may involve operations such as edge detection, morphological filtering, and candidate region analysis based on geometric constraints such as aspect ratio and size.

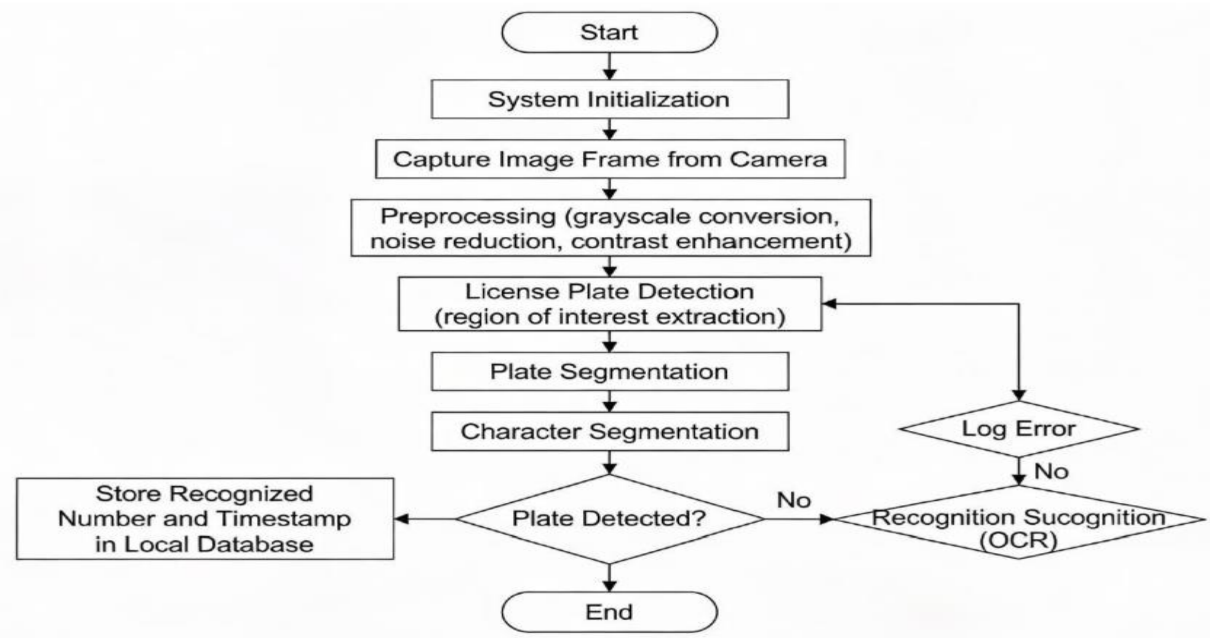

Fig. 3.4. Flowchart of ANPR algorithm

In the subsequent stage, character segmentation is performed on the extracted plate region. This step separates individual characters using techniques such as thresholding, connected component analysis, and projection profiles. Each segmented character image is then normalized to a standard size and passed to the OCR engine. The OCR module uses computer vision and pattern recognition techniques to map the character images to corresponding alphanumeric symbols, thereby reconstructing the complete number plate string.

3.5 Power Management and Solar Integration Strategy

The power management strategy is designed to ensure reliable continuous operation of the ANPR system under varying solar and load conditions. The overall solar power management architecture is illustrated in fig. 3.6.

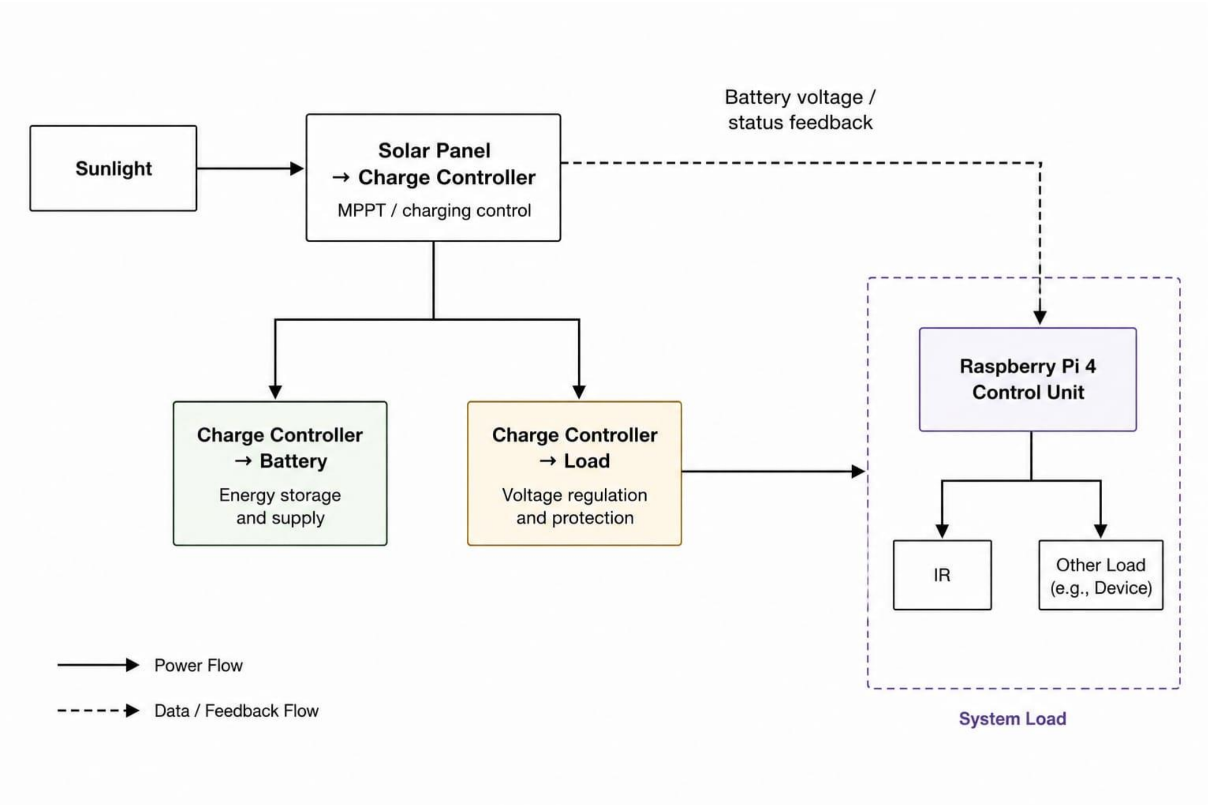

Fig. 3.5. Solar power management architecture

During daylight hours, the solar panel generates electrical power that is regulated by the charge controller, which prioritizes charging the battery while simultaneously supplying the system load if sufficient energy is available. The charge controller implements protection mechanisms such as overcharge, short-circuit, and reverse polarity protection to enhance system safety and battery life.

Power management policies are implemented at the system level to optimize energy usage. For example, the Infrared (IR) illumination is activated only when ambient light falls below a defined threshold, thereby reducing unnecessary power consumption during daytime operation. Similarly, image capture and processing rates can be adjusted according to application requirements and available battery charge, allowing the system to operate in a low-power mode during extended periods of poor solar input.

In addition, the system monitors key electrical parameters such as battery voltage and panel voltage and current, enabling basic diagnostics and preventive actions. If the battery voltage drops below a critical threshold, the system can enter a safe shutdown or reduced functionality mode to prevent deep discharge and to preserve battery health.

3.6 System Workflow

The overall system workflow combines the hardware and software subsystems into a coherent operational sequence, as depicted in fig. 3.6.

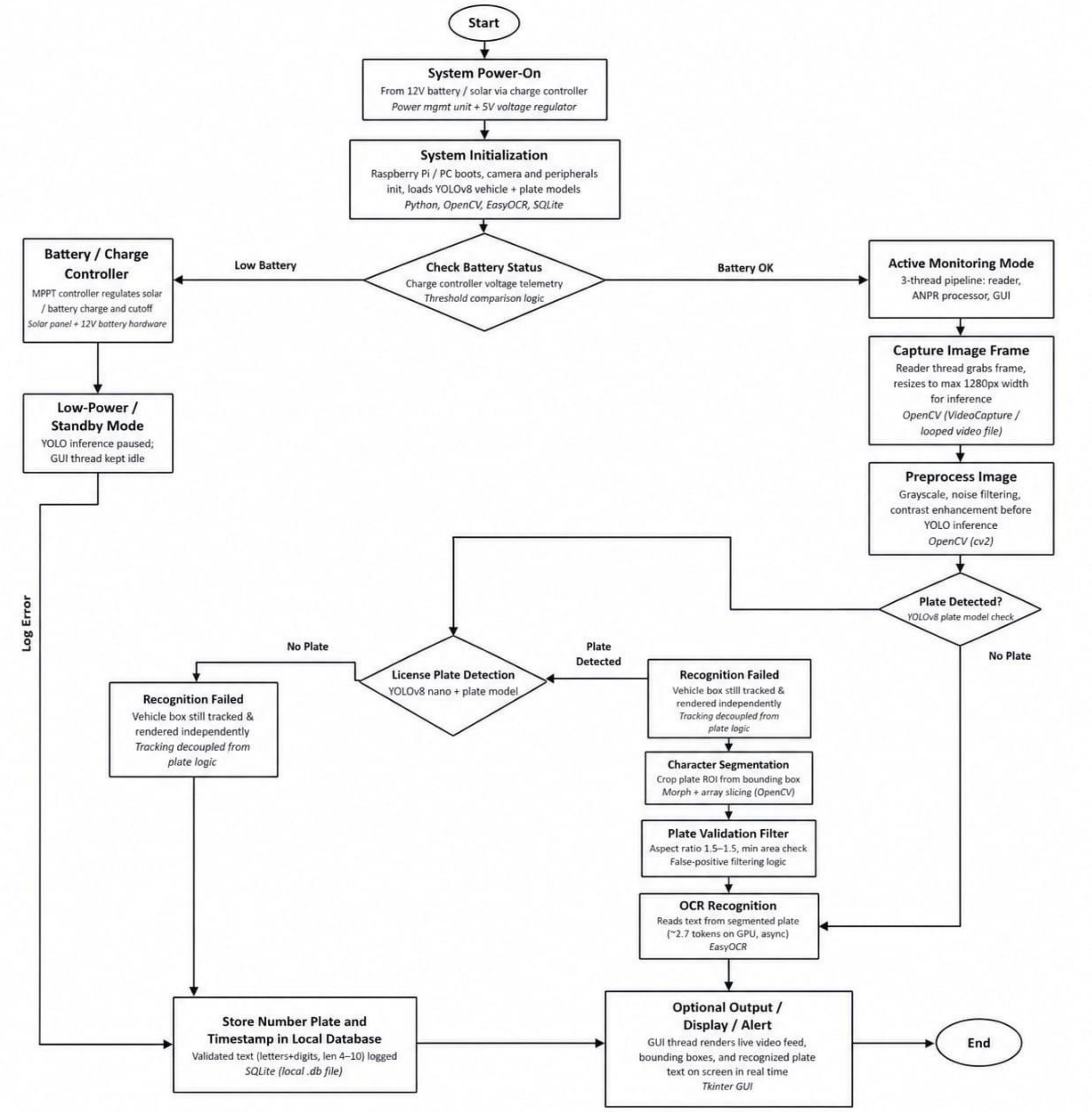

Fig. 3.6. Overall system workflow

When power becomes available from the battery, the system powers up and performs an initialization routine, which includes configuring the camera, setting up the ANPR algorithm parameters, and verifying the status of the solar and battery subsystems. In the active monitoring state, the camera periodically captures frames of the vehicle lane according to the configured sampling rate.

For each captured frame, the software checks whether the scene contains a vehicle and a visible number plate; if not, the frame is discarded and the system returns to the capture stage. When a valid candidate is detected, the frame is passed through the preprocessing and plate localization stages. If the plate is successfully localized, the plate region is extracted, segmented, and recognized using the OCR module to obtain the vehicle registration number. The recognized number plate along with a timestamp is then stored in non-volatile memory or transmitted to a local interface.

Throughout the workflow, the system periodically checks the battery status and power input conditions. If the battery voltage falls below a predefined threshold, the system transitions to a low-power or standby mode, where non-essential functions are reduced or temporarily disabled. When the battery level recovers due to sufficient solar charging, the system resumes normal operation automatically.

3.7 Algorithm: Automatic Vehicle Number Plate Recognition (ANPR) System

Step 1: Start the system.

Step 2: Initialize the Raspberry Pi, camera module, IR illuminator, and required software libraries.

Step 3: Check the battery status. If the battery level is below the threshold, switch the system to standby mode; otherwise, continue.

Step 4: Capture image frames continuously from the camera.

Step 5: Preprocess the captured image by applying resizing, grayscale conversion, noise reduction, and contrast enhancement.

Step 6: Detect the vehicle and locate the license plate using the YOLOv8 model.

Step 7: If no license plate is detected, log the event and capture the next frame.

Step 8: If a license plate is detected, crop the plate region.

Step 9: Segment the characters from the cropped license plate image.

Step 10: Recognize the characters using the OCR engine.

Step 11: Validate the recognized license plate number.

Step 12: Store the recognized license plate number along with the timestamp in the local database.

Step 13: Display the detected vehicle, license plate, and recognition result on the graphical user interface.

Step 14: Repeat Steps 4–13 until the system is stopped.

Step 15: End.

3.8 Methodology Summary

This chapter has presented the methodology adopted for the design and implementation of the proposed automatic vehicle number plate recognition system using computer vision techniques powered by a solar energy subsystem. The overall system architecture was introduced and represented through a functional block diagram, illustrating the interaction between the camera, Infrared (IR) illumination, processing unit, storage, and solar power supply.

The hardware design and component selection emphasized a balance between computational capability, power efficiency, and environmental robustness, while the software architecture was developed as a structured ANPR pipeline including image acquisition, preprocessing, plate localization, character segmentation, and OCR-based recognition. The combined system workflow described how these hardware and software elements interact during real-time operation to capture, process, and store number plate information under varying power and environmental conditions.

CHAPTER 4: RESOURCE UTILIZATION

The following table presents the component-wise bill of materials along with the estimated cost for the proposed ANPR prototype system.

Item	**Quantity**	**Price / Unit (₹)**	**Total (₹)**
High-Resolution Camera Module	1	1,500	1,500
Raspberry Pi 4	1	2,500	2,500
Infrared (IR) Illuminator / Night Vision LED	1	400	400
64 GB microSD Card / SSD Storage	1	600	600
Power Supply / Adapter (12V–5V)	1	300	300
Weatherproof Housing / Mount	1	500	500
Wi-Fi / 4G Network Module	1	700	700
Cloud Server / Hosting (Annual)	1	1,200	1,200
Miscellaneous Components (Cables, Connectors)	1	300	300
Grand Total			**₹ 8,000**

CHAPTER 5: MEASUREMENTS, RESULTS AND DISCUSSION

5.1 Experimental Setup

The experimental setup of the proposed automatic number plate recognition (ANPR) system consists of a raspberry pi 4, an Infrared (IR) night vision camera module, a solar power supply unit, and the required software environment for image processing and number plate recognition. The complete setup is designed as a standalone embedded platform capable of performing vehicle image acquisition and number plate detection using computer vision techniques.

The raspberry pi 4 is used as the central processing unit of the system. It is responsible for image acquisition, preprocessing, number plate localization, optical character recognition (OCR), and data storage operations. The raspberry pi platform was selected because of Intelligent Transportation Systems (ITS) compact size, low power consumption, and compatibility with python-based computer vision libraries.

An Infrared (IR) night vision camera module is integrated with the raspberry pi to capture vehicle images under both daytime and nighttime conditions. The camera is positioned in such a way that vehicle number plates remain clearly visible within the field of view. Infrared illumination support enables image capture even in low-light environments without causing visible glare.

The software implementation is carried out using python programming language along with OpenCV and OCR libraries. OpenCV is used for image preprocessing, edge detection, contour analysis, and number plate localization, while OCR techniques are used for extracting alphanumeric characters from the detected number plate region.

A solar-powered energy subsystem is also incorporated into the setup to provide uninterrupted operation of the system. The solar power unit consists of a solar panel, rechargeable battery, and voltage regulation circuitry. This subsystem is intended to support standalone operation in remote or power-scarce areas where continuous grid electricity may not be available.

Fig. 5.1. Experimental hardware setup of the proposed system

5.2 Software Development Progress

The software development process of the proposed Automatic Number Plate Recognition (ANPR) system is currently in the implementation and testing phase. The system software is being developed using python programming language on the raspberry pi platform. Various computer vision and image processing libraries have been installed and configured successfully for developing the number plate recognition pipeline.

The initial stage of software development involved configuring the raspberry pi operating environment and establishing communication with the Infrared (IR) night vision camera module. Successful image acquisition tests were performed to verify proper camera interfacing and image capture functionality under different lighting conditions.

OpenCV libraries are being utilized for performing image preprocessing operations such as grayscale conversion, noise reduction, edge detection, thresholding, and contour analysis. These preprocessing operations are essential for improving the visibility and localization of vehicle number plates from captured images.

The OCR module is being integrated for automatic extraction of alphanumeric characters from the detected number plate region. Preliminary OCR testing has been performed on sample images to evaluate character recognition capability and processing performance. The software development process also includes implementation of data logging and storage mechanisms for saving recognized number plates along with captured images and timestamps.

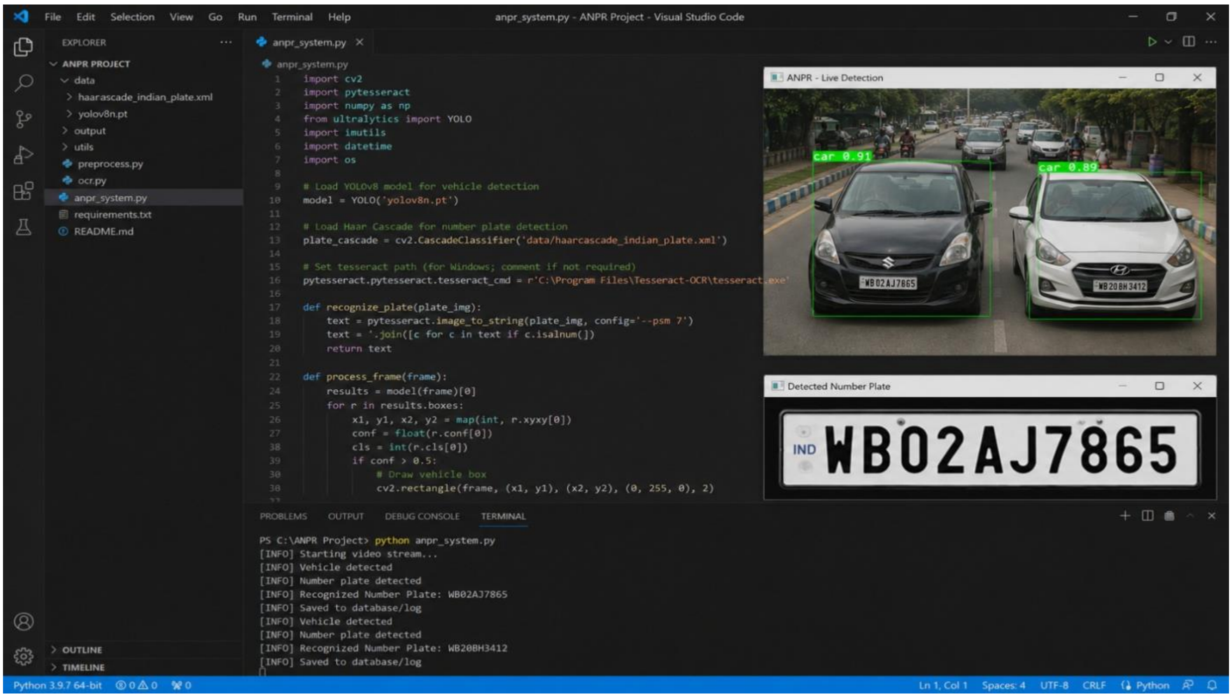

Fig. 5.2. Software development and testing environment

5.3 Preliminary Testing

Preliminary testing of the proposed automatic number plate recognition (ANPR) system has been conducted to evaluate the basic functionality of the hardware and software modules. Initial testing was mainly focused on verifying image acquisition, preprocessing operations, and basic number plate localization capability using the raspberry pi platform and Infrared (IR) night vision camera module.

During the testing process, vehicle images were captured under different environmental and lighting conditions. The captured images were processed using opencv-based image processing techniques such as grayscale conversion, image enhancement, edge detection, and contour analysis. These operations were performed to improve image quality and identify possible number plate regions from the captured vehicle images.

The preliminary testing results indicate that the system is capable of successfully capturing vehicle images and performing basic preprocessing operations required for number plate recognition. Initial observations also suggest that the Infrared (IR) camera module improves visibility during low-light conditions and supports image acquisition during nighttime operation. The OCR module is currently under further testing and optimization for improving character recognition performance.

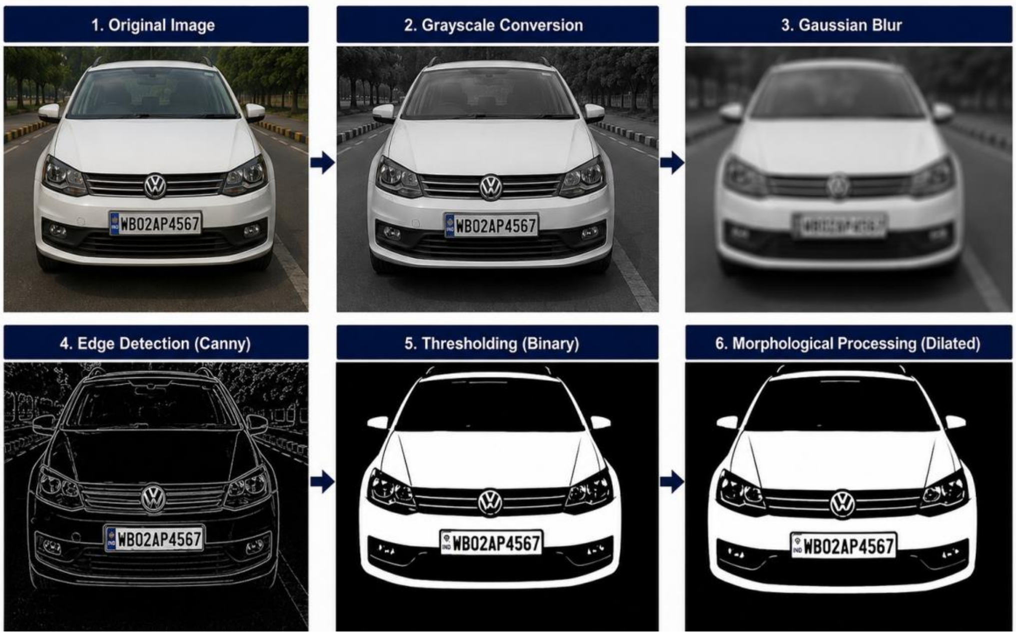

Fig. 5.3. Preliminary testing and image processing output

5.4 Expected System Output

The expected output of the proposed system includes automatic detection and recognition of vehicle number plates from captured images using computer vision and OCR techniques. The system is designed to identify the number plate region from the vehicle image, extract the alphanumeric characters, and generate the recognized registration number automatically.

The processing pipeline begins with image acquisition using Infrared (IR) camera module. The captured image is then preprocessed using OpenCV techniques such as grayscale conversion, noise filtering, thresholding, and edge detection. After preprocessing, the system attempts to localize the number plate region using contour-based analysis and geometric filtering methods.

Once the number plate region is successfully detected, OCR techniques are applied to recognize and extract the alphanumeric characters present on the number plate. The recognized output is expected to be stored along with the captured image and timestamp information for future monitoring and analysis purposes.

Fig. 5.4. Vehicle number plate detection output

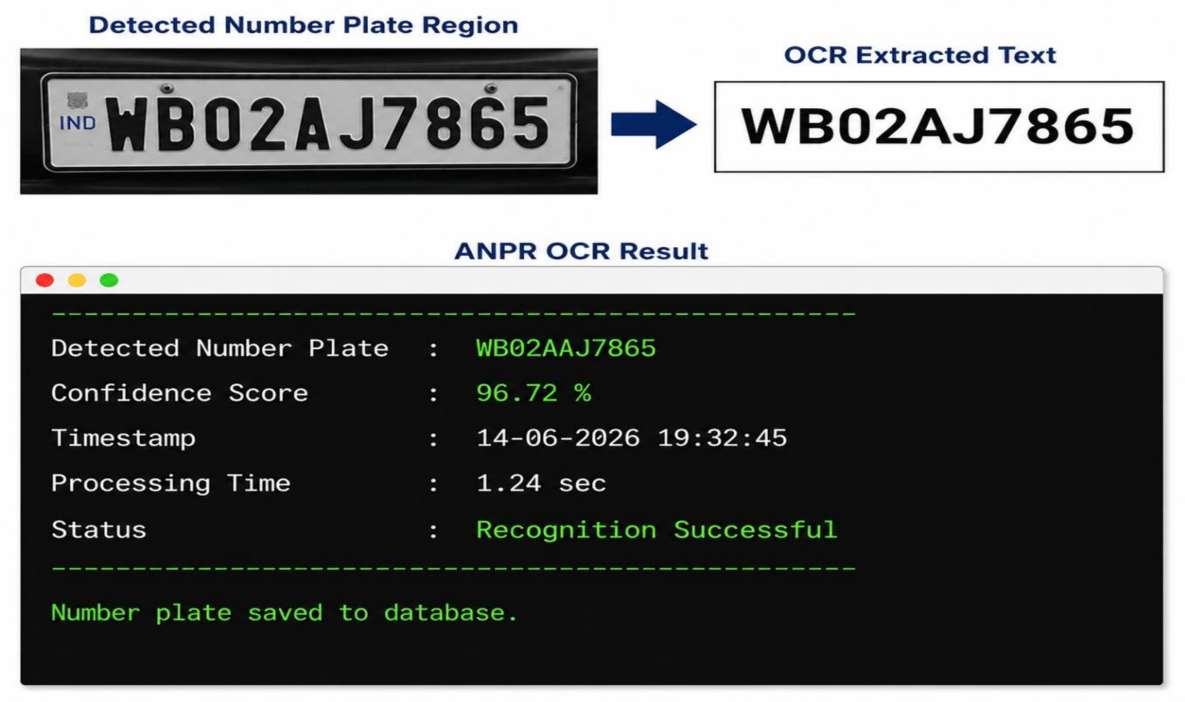

Fig. 5.5. OCR-based number plate recognition output

5.5 Challenges Faced During Development

Several practical and technical challenges were encountered during the development phase of the proposed ANPR system. One of the major challenges was achieving accurate number plate detection under varying lighting conditions, especially during nighttime operation and low ambient illumination. Variations in image quality and reflections from vehicle surfaces also affected the detection performance during initial testing.

Another significant challenge involved optimizing the OCR module for different vehicle number plate formats, fonts, and image conditions. In some cases, image noise, motion blur, and improper camera alignment reduced character recognition accuracy and required additional preprocessing and optimization techniques.

The integration of the Infrared (IR) camera module with the raspberry pi platform also required careful configuration and testing to ensure stable image acquisition and proper synchronization with the image processing pipeline. Power management and integration of the solar-powered subsystem presented additional challenges related to voltage regulation, battery backup, and continuous operation of the system.

CHAPTER 6: CONCLUSION AND SCOPE OF FUTURE WORK

6.1 Conclusion

This project presents the design and development of an automatic vehicle number plate recognition (ANPR) system using computer vision techniques on a raspberry pi-based embedded platform. The proposed system integrates Infrared (IR) night vision camera, OCR techniques, python-based image processing, and solar-powered operation for automated vehicle monitoring and number plate recognition applications.

The project focuses on developing a low-cost, energy-efficient, and portable ANPR solution capable of operating under both daytime and nighttime conditions. Initial implementation and preliminary testing demonstrate that the proposed architecture is capable of image acquisition, preprocessing, and basic number plate localization operations using opencv techniques.

The integration of embedded processing and computer vision methods provides a practical approach for intelligent transportation, surveillance, and vehicle monitoring applications. Although the project is currently in the development and testing phase, the implemented system architecture establishes a strong foundation for further optimization and real-time deployment. The proposed system can potentially be used in applications such as traffic monitoring, parking management, toll collection, security surveillance, and automated vehicle access control systems.

6.2 Scope of Future Work

Several improvements and advanced features can be incorporated into the proposed system in future development stages. Further optimization of image preprocessing and OCR algorithms can improve the overall number plate detection and recognition accuracy under different environmental conditions.

Machine learning and deep learning-based approaches may be integrated in future versions of the system for enhanced vehicle detection and intelligent character recognition. Cloud-based data storage and wireless communication modules can also be added for real-time monitoring and centralized database management.

Future work may also include implementation of automatic alert generation, vehicle database verification, and integration with smart city infrastructure for intelligent traffic management applications. Additional testing and optimization are expected to improve system stability, processing speed, and operational reliability in real-world conditions. The exploration of blockchain technology for secure, immutable record-keeping represents another promising avenue for future research.

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